

Assembly & Wiring Instruction Sheet

This document organizes the build process into construction, soldering, and software installation stages. Use the checkboxes to track progress.

1. Construction / Printing & Assembly

- ☐ 1. Print core upper and lower components.
 - ☐ 2. Print reservoir and back frames.
 - ☐ 3. Print front extended or normal frames.
 - ☐ 4. Print FED3 holder for frame.
 - ☐ 5. Print 3× lickports / frames / CS frames.
 - ☐ 6. Print floor pan (extended or normal).
 - ☐ 7. Print swing arms.
 - ☐ 8. Print acrylic rims.
 - ☐ 9. Print solenoid holder.
 - ☐ 10. Press-fit M3/M4 nuts into core frame. COMPLETE SOLDERING BEFORE CONTINUING
-
- ☐ 11. Install Arduino, relay, and solenoid frame onto core base plate using M3/M4 screws.
 - ☐ 12. Install FED3 aux cable into GND / D5 screw terminals.
 - ☐ 13. Plug 12-pin cable (3× 5-pin cables with each losing 1 pin to become 3× 4-pin cables) into CS PWM female connectors.
 - ☐ 14. Install tubing from syringe barrels to middle of Leco solenoid, opposite terminal wire port, to lickport luer adapter.

2. Soldering / Wiring

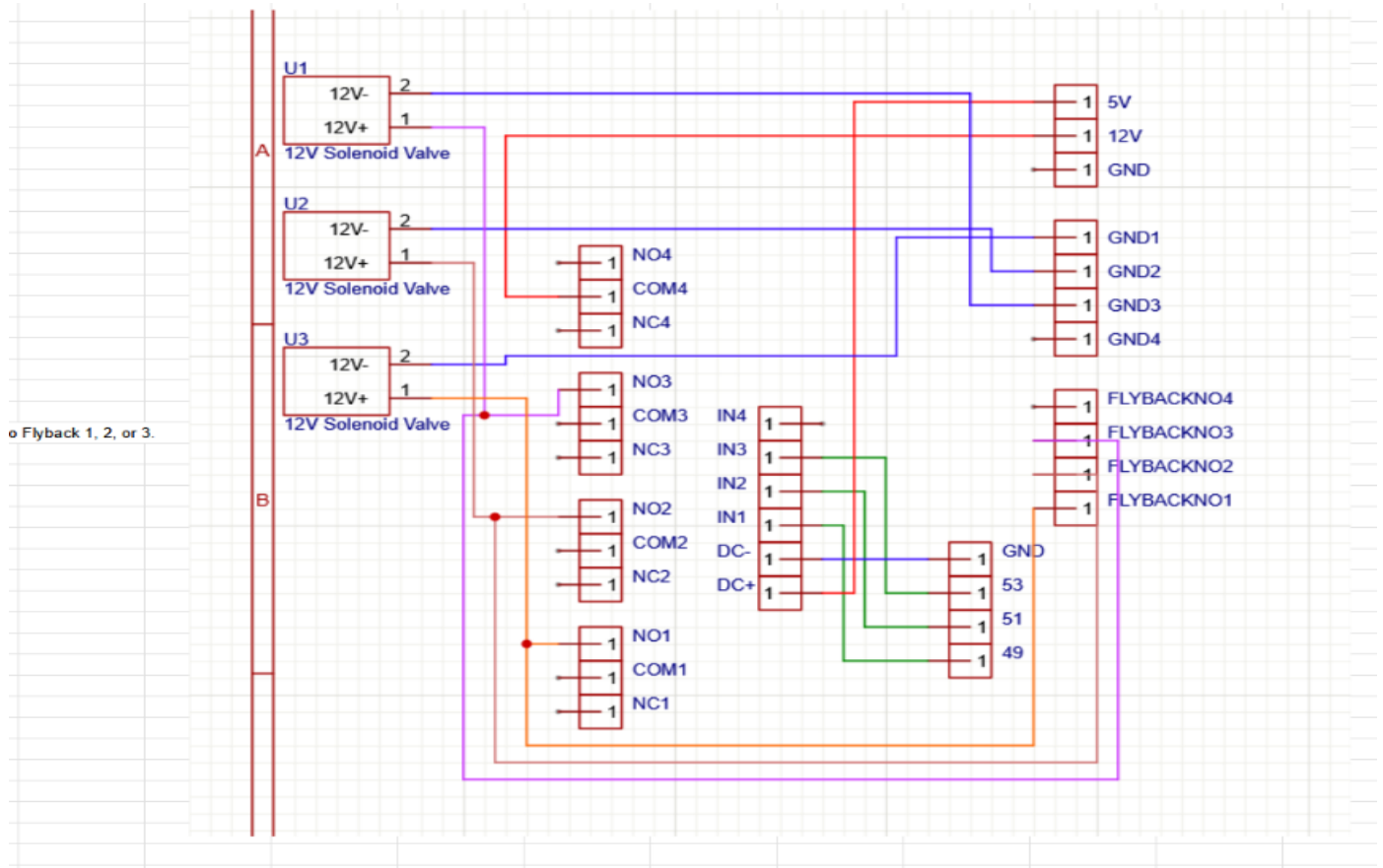
- ☐ 1. Solder diodes and resistors (1k–1M ohm).
- ☐ 2. Solder Arduino pin connections (male).
- ☐ 3. Solder ESP32 pin connections (female).
- ☐ 4. Solder BNC, ESP32 D5, CS GND, lickport, relay, and relay-trigger screw terminals.
- ☐ 5. Solder Arduino CS PWM connections (female).
- ☐ 6. Solder Micro SD card, RTC, and LCD screen.

- ☐ 7. Tin 5 wires and solder onto BNC panel mount 1–5.
- ☐ 8. Tin 4 small wires and 1 long wire. Solder daisy-chain BNC grounds together and plug into core GND.
- ☐ 9. Solder all COM terminals together underneath relay module.
- ☐ 10. Tin and solder 5V, 1 GND, 3 signal wires, and install 5V, GND, 49, 51, and 53 to relay.
- ☐ 11. Tin 9 wires and solder from solenoid GND to PCB GND.
- ☐ 12. For remaining 6 wires, use 20 AWG. Twist one end together in parallel to create 3 wire sets with a 'V' connection. Solder one end to solenoid 12V and connect the V into relay NO. Connect remaining ends into Flyback 1, 2, or 3.
- ☐ 13. Tin one additional wire and attach from PCB 12V to COM4. (Power path: 12V → COM → NO (on signal) → Solenoid+ → Solenoid– → PCB GND. Flyback diode NO to GND.)
- ☐ 14. Solder 9× 240Ω resistors onto 3 CS module PCBs.
- ☐ 15. Solder 3× male 5-pin headers onto the back of the CS PCBs.
- ☐ 16. Solder 3× RGB LEDs onto front of CS module, ensuring orientation is correct.
- ☐ 17. Solder 3× buzzer onto front of CS module (5-pin).
- ☐ 18. For CS wiring: connect 3 shielded wires together at the male GND and extend to PCB GND. (Requires converting 5-pin to 4-pin by separating GND and creating space on PWM row.)

3. Software Installation

- ☐ 1. Flash ESP32 with ESP32 setclock / logging scripts (set clock, then reflash with logging).
- ☐ 2. Flash Arduino with required control script.

Standalone Wiring Schematic



Reference schematic for relay, solenoid, flyback diode, and PCB wiring connections.